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A Band-Matching Descriptor Breaks Scaling Relations for Sulfur Electrocatalysts

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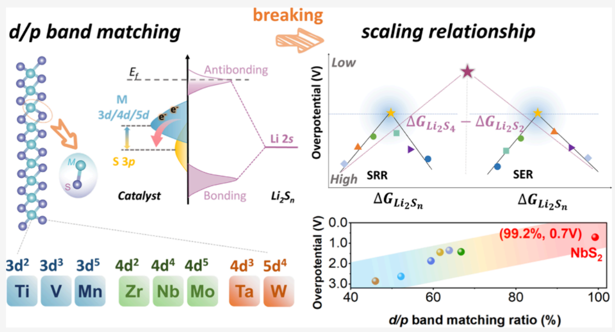
ABSTRACT: A predictive descriptor is crucial for the rational design of catalysts that enable fast and reversible sulfur redox in lithium−sulfur (Li−S) batteries. Although several descriptors have been proposed, most remain constrained by the conventional adsorption−activity trade-off and the associated volcano-type behavior. Here, we uncover that the catalytic activity of widely used transition-metal sulfides is dictated by the *d/p* band-matching ratio between metal centers and surface sulfur atoms, thereby going beyond this limitation. By tuning the band-matching ratio, which modulates the interaction between lithium polysulfides and the catalyst surface, a linear correlation with the overpotentials of both sulfur reduction and evolution reactions is established, accelerating the sulfur conversion kinetics. NbS_2, with a band-matching ratio of 99.2%, delivers a minimal bifunctional overpotential of 0.70 V (the sum of sulfur reduction reaction (SRR) and sulfur evolution reaction (SER) overpotentials), 4-fold lower than that of WS_2 (2.85 V, with a band-matching ratio of 46.0%). The Li−S battery that delivers a high initial areal capacity of 10.98 mAh cm^{-2} with the NbS_2 catalyst exhibits an exceptionally high retention of 90.29% after 100 cycles, surpassing most of the reports. This work establishes band matching as a general principle for sulfur electrocatalysis, providing a predictive design rule that transcends classical scaling relations for metal−sulfur batteries.



INTRODUCTION

The scaling relationship, initially developed to estimate hydrogenation and dehydrogenation reaction energies, has evolved into a universal framework in heterogeneous catalysis for predicting intermediate binding energies.[1,2] By further integrating with the Brønsted−Evans−Polanyi relation linking activation barriers and reaction energies, the complete potential energy diagrams for transition-metal-catalyzed surface reactions can be determined, thereby constituting a powerful tool for screening novel catalysts in multielectron and multiphase transfer reactions.[3] A prototypical case is sulfur electrochemistry in batteries, which features a 16-electron transfer mechanism and complex solid−liquid−solid phase transitions. They have attracted intense interest for their potential to deliver ultrahigh energy densities in next-generation batteries for electric vehicles and related applications. Consequently, sulfur electrocatalysis has emerged as a vibrant research field focused on fundamentally overcoming the kinetic bottlenecks of sulfur conversion reactions and suppressing the shuttle effect.[4−7]

Based on the bond-order conservation theory, when a catalyst exhibits strong *d-p* or *s-p* orbital coupling with a particular intermediate species, it generally demonstrates comparable adsorption strength toward another intermediate, yielding intrinsic linear correlations that simplify screening. For multielectron transfer reactions in lithium−sulfur (Li−S) batteries, when such a relationship holds, it allows for the prediction of the theoretical upper limit on catalytic activity (Figure 1a).[8,9] In contrast, the absence of a linear correlation enables tuning and optimization of the catalytic performance by modulating the binding strengths of the key intermediate. However, comprehensive mechanistic investigations capable of verifying whether a linear scaling relationship exists among the binding energies of various polysulfide intermediates are still lacking. This gap severely hinders the rational design and optimization of efficient electrocatalysts for Li−S batteries. Consequently, elucidating the electronic-structure origins of these scaling relationships and developing strategies to break them are critical for achieving breakthroughs in sulfur electrocatalysis.[10,11]

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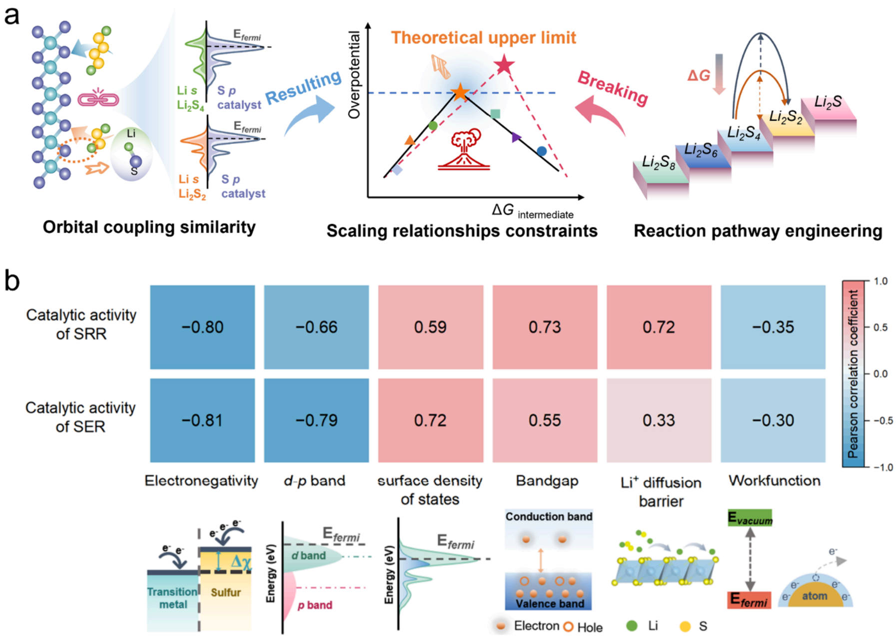

Figure 1. Scaling relationships originating from orbital coupling govern catalytic activity in Li−S chemistry. (a) Orbital-derived linear correlation of adsorption energies among LiPSs limits activity optimization. (b) Pearson correlation analysis of catalytic activity with multiple electronic descriptors for sulfur reduction and oxidation reactions.

Overcoming such intrinsic scaling limitations in complex multielectron systems requires directly targeting the fundamental electronic features of the reaction pathway constraints. Recent studies reveal that constructing a Ni−Fe_2 dual-site molecular catalyst enables dynamic evolution of the Ni site, which effectively modulates the electronic structure of adjacent Fe active centers, thereby breaking the scaling relationship in the oxygen evolution reaction. Meanwhile, a fluorine-doped carbon-supported Pt catalyst was developed, in which interfacial polarity and a superhydrophobic microenvironment decouple the *OOH/*OH adsorption energy scaling relation in oxygen reduction.[12−20] However, unlike oxygen electrocatalysis involving 2-electron or 4-electron transfer pathways, sulfur electrocatalysis features a more complex reaction network with diverse intermediates (Li_2S_n, $2 \leq n \leq 8$). Such complexity is further compounded by the inherent chemical instability of polysulfide intermediates. Specifically, at concentrations relevant to battery operation, nominal Li_2S_4 undergoes spontaneous disproportionation into soluble Li_2S_6 and solid Li_2S.[21−23] Merely modulating the single electronic structure of the key intermediate is insufficient to break the scaling relationship in Li−S batteries.[24,25] The most direct strategy of breaking the scaling relationship is achieved through pathway engineering focused on the rate-determining step (RDS), which essentially requires the regulation of multiple electronic structures, including band structure, atomic electronegativity, work function, ionic conductivity, surface density of states, and so on (Figure 1b). As the RDSs govern both the overpotential and the kinetic bottlenecks, constructing a descriptor that quantitatively reflects the energy difference between these critical steps and correlates with the catalyst's electronic structure is essential for breaking scaling relationships through pathway-level mechanistic regulation.

In this work, we introduce a d/p band-matching ratio, which is defined by the electronic states of metal centers and surface sulfur atoms in layered transition-metal sulfides (TMSs), as a predictive descriptor for sulfur electrocatalysis. We show that this band-matching ratio exhibits a linear correlation with the overpotentials of both sulfur reduction and sulfur evolution, enabling quantitative ranking across TMS catalysts. This descriptor captures the coupled effects of the intrinsic electronic configuration of the metal center and the degree of metal−sulfur *d-p* alignment (quantified by the separation of their band-center energies), providing a pathway-dependent metric that goes beyond the conventional adsorption−activity volcano paradigm. Optimizing band matching enhances the electronic density at surface sulfur sites and facilitates interfacial charge transfer, thereby lowering the kinetic barriers to sulfur conversion. Guided by this descriptor, NbS_2 (band-matching ratio of 99.2%) is identified as an optimal TMS catalyst, enabling Li−S cells to cycle stably for 1,000 cycles with a decay rate of 0.086% per cycle. Even under a high sulfur loading (9.89 mg cm^{-2}), 90.29% of their areal capacity is retained after 100 cycles. Overall, these results establish band-matching as a unified design principle to circumvent scaling constraints in metal−sulfur batteries.

RESULTS AND DISCUSSION

Revealing the Scaling Relationship

TMSs have been extensively explored in conventional industrial and Li−S battery catalysis.[26−29] Layered TMSs from Groups IVB to VIIB were selected as model catalysts, including TiS_2, VS_2, MnS_2, ZrS_2, NbS_2, MoS_2, TaS_2, and WS_2. HfS_2, despite being a typical layered TMS, possesses a relatively large band gap (∼1.96 eV), indicating poor electronic conductivity, and was therefore excluded.[30,31] TcS_2 was not considered due to the radioactive nature of technetium and the limited synthetic accessibility of Tc-based compounds, which makes it impractical for catalyst design.[32] CrS_2 was also excluded because of its excessively strong Li_2S adsorption energy (5.24 eV), high work function (∼7.03 eV), and potential biological toxicity, all of which are unfavorable for reversible polysulfide conversion (Figure 2a).[33−35] In theoreti-

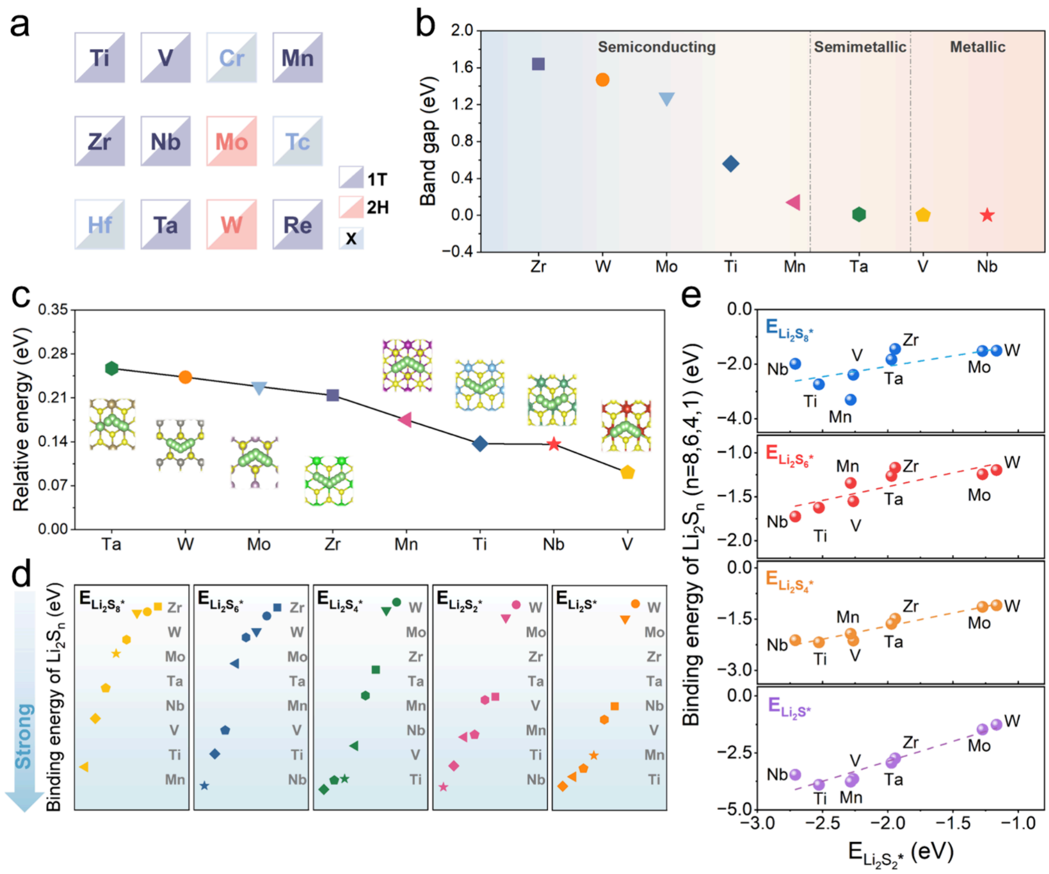

Figure 2. Electronic structure, lithium-ion transport properties, and adsorption characteristics of TMSs. (a) Catalysts investigated in this study. (b) Calculated bandgaps of TMSs. (c) Lithium-ion migration energy barriers in TMSs. (d) Computational results of TMS adsorption capacities for various sulfur species. (e) Scaling relationship among adsorption energies of sulfur species.

cal calculations, the primary step entails the identification of the thermodynamically most stable crystalline phase of the catalyst. Among all TMSs, only MoS_2 and WS_2 exhibit the stable 2H phase, while the others adopt the 1T phase (Figure S2).[36] Furthermore, the catalytically active exposed surfaces of all TMSs correspond to the (001) crystal plane. Electrical conductivity and lithium-ion conductivity are the prerequisites for catalyst selection in batteries.[37,38] The generalized gradient approximation with the Perdew−Burke−Ernzerhof (GGA-PBE) method was employed to calculate the band structures (Figure S3) and density of states (DOS) of the TMS (Figure S4). Figure 2b shows that small band gaps (0−1.64 eV) are generally present between the valence band maximum (VBM) and the conduction band minimum (CBM) of TMSs. Notably, NbS_2 and VS_2 exhibit a band gap of 0 eV, demonstrating excellent electrical conductivity. The migration ability of lithium ions was calculated using the climbing image nudged elastic band (CI-NEB) method.[39] For all TMSs, the maximum lithium-ion diffusion energy barrier does not exceed 0.26 eV, demonstrating excellent lithium-ion conductivity (Figures 2c and S6).

A systematic computational analysis incorporating van der Waals corrections (DFT-D3) was then conducted to evaluate their anchoring ability for polysulfides, and a graphene layer with a (30 × 30 × 20) Å^3 supercell was built as a comparison. As shown in Figure S7, lithium polysulfides (LiPSs) are adsorbed on the TMS surface via Li−S bonds, with bond lengths ranging from 2.37 to 2.89 Å. The binding energies are all below −3.90 eV, indicating moderate adsorption strength between LiPSs and TMSs (Figure 2d).[40] Elucidating the relationships between the binding energies of polysulfides is crucial for regulating the catalytic activity. As shown in Figure 2e, the binding energies of Li_2S_8, Li_2S_6, Li_2S_4, and Li_2S exhibit linear correlations with that of Li_2S_2, where the increased binding energy of Li_2S_2 is accompanied by proportionally enhanced binding strengths of other polysulfide intermediates. This scaling relationship highlights intrinsic constraints among polysulfide adsorption behaviors on TMS surfaces.[2] Besides, a recent study on the 3*d* transition metals into the nitrogen-doped defective black phosphorus carbide catalyst demonstrated strong adsorption energy scaling relations among sulfur intermediates, with correlation coefficients of about 0.9.[41] This also indicates that modifying the binding energy of one intermediate will concurrently influence the binding energies of other LiPSs. Thus, solely regulating the adsorption strength of a single intermediate cannot regulate the conversion route to realize a high catalytic activity.

Catalytic Activity Regulation Inspired by the Scaling Relationship

Inspired by oxygen electrocatalysis, in which the Gibbs free energy of an elementary reaction pathway (e.g., ΔG_{OOH*} and ΔG_{O*}) serves as a critical descriptor for reaction overpotential, we thus propose that establishing a correlation between the reaction overpotential and the Gibbs free energy difference associated with the LiPS conversion pathway could be an effective approach to break the scaling relationship. The overpotential (Figure 3a) derived from Gibbs free energy for

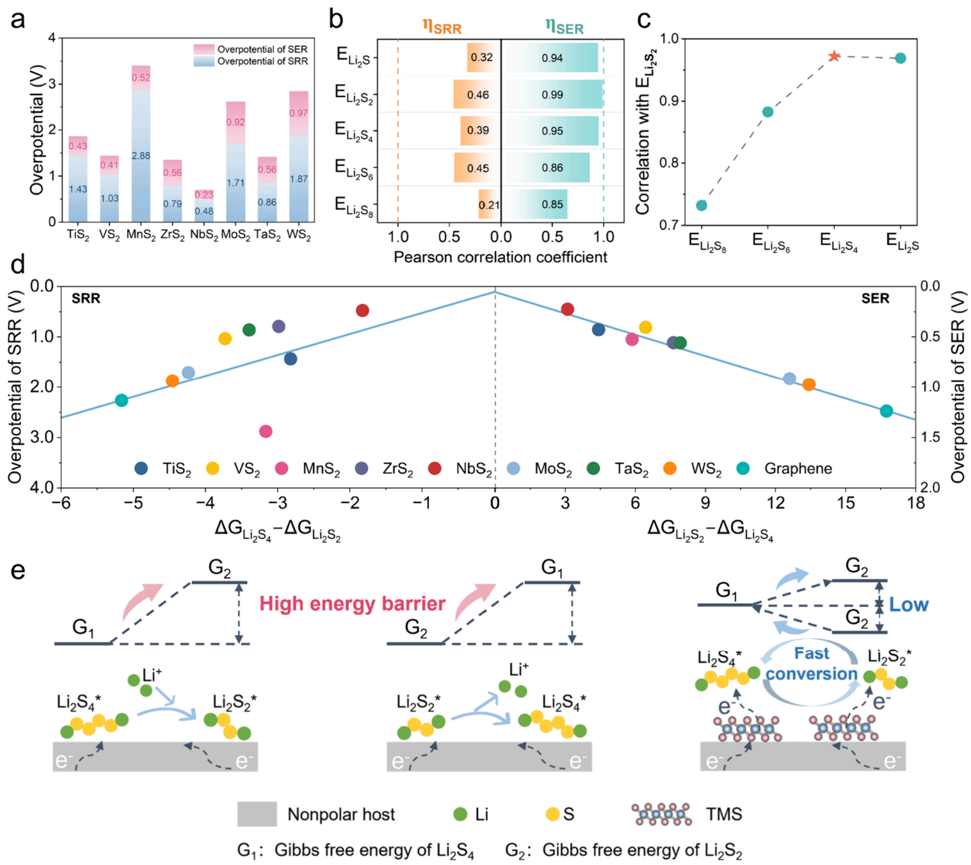

Figure 3. Correlating polysulfide adsorption and catalytic activity of TMSs: toward breaking scaling relationships through conversion-pathway regulation. (a) Theoretical overpotentials of TMSs for SRR and SER. (b) Pearson correlation between theoretical overpotentials and binding energies of sulfur species. (c) Pearson correlation between the adsorption energies of other sulfur species and those of Li_2S_2. (d) Linear relationship between overpotential and the Gibbs free energy difference for both the liquid−solid conversion ($Li_2S_4 \rightarrow Li_2S_2$) in the SRR and the corresponding solid−liquid conversion ($Li_2S_2 \rightarrow Li_2S_4$) in the SER. (e) Mechanistic illustration of conversion-pathway regulation to overcome scaling constraints in SRR and SER.

each conversion step of SRR and sulfur evolution reaction (SER) (Figures S8 and S9) shows that compared to the relatively high SRR overpotential (2.27 V) and SER overpotential (1.24 V) of carbon-based materials, the overpotential of TMS systems is significantly decreased.[42,43] Among TMSs, NbS_2 exhibits the lowest overpotentials for both SRR (0.48 V) and SER (0.23 V), indicating its superior catalytic activity.[44] Specifically, NbS_2 reduces the energy barrier of the liquid−solid conversion process of Li_2S_4 to Li_2S_2 and shifts the rate-determining step of SRR to the subsequent solid−solid conversion step of Li_2S_2 to Li_2S, effectively enhancing sulfur conversion efficiency. Furthermore, NbS_2 significantly lowers the oxidation reaction barrier of the Li_2S/Li_2S_2 in the SER. In contrast, other TMSs maintain their rate-determining steps at earlier stages (e.g., the Li_2S_4−Li_2S_2 conversion process), resulting in notably higher overpotentials.

The LiPS adsorption energies have been widely recognized as a key factor affecting the overpotential. Figure 3b shows the correlation between the binding energy of LiPSs and the overpotential of SRR and SER. For SRR, the binding energy of individual polysulfides exhibits insignificant correlations with overpotential, with all calculated coefficients below 0.5.[45] For the SER, all LiPS binding energies exhibit significant correlations with overpotential, with Li_2S_2 showing the strongest correlation (R = 0.99). These suggest that the catalytic activity of SRR and SER cannot be effectively regulated or predicted by the adsorption strength of a single intermediate. This finding corroborates the constraints imposed by the scaling relationship among binding energies of LiPSs proposed in the preceding section.

Pearson correlation analysis revealed that among all LiPSs, the binding energies of Li_2S_4 and Li_2S_2 exhibit the highest correlation of 0.97 (Figure 3c). Consequently, we correlated the reaction overpotentials with the Gibbs free energy differences for both the liquid−solid conversion ($Li_2S_4 \rightarrow Li_2S_2$) in SRR and the corresponding solid−liquid conversion ($Li_2S_2 \rightarrow Li_2S_4$) in SER. These two processes precisely match the experimentally identified rate-determining steps, confirming their roles as intrinsic energetic bottlenecks that simultaneously constrain reduction and oxidation reactions. By explicitly associating catalytic activity with these rate-determining conversion pathways, we successfully proposed the linear relationship observed in Figure 3d. In this relationship, a decreasing Gibbs free energy difference corresponds to enhanced catalytic activity with NbS_2 located at the optimal point. This pathway-dependent descriptor effectively couples the sulfur reduction and oxidation processes, enabling a unified assessment of catalytic performance across the full redox pathway. Therefore, the linear relationship provides clear theoretical guidance: narrowing the Gibbs energy difference between the rate-determining conversion pathway flattens the reaction energy profile,

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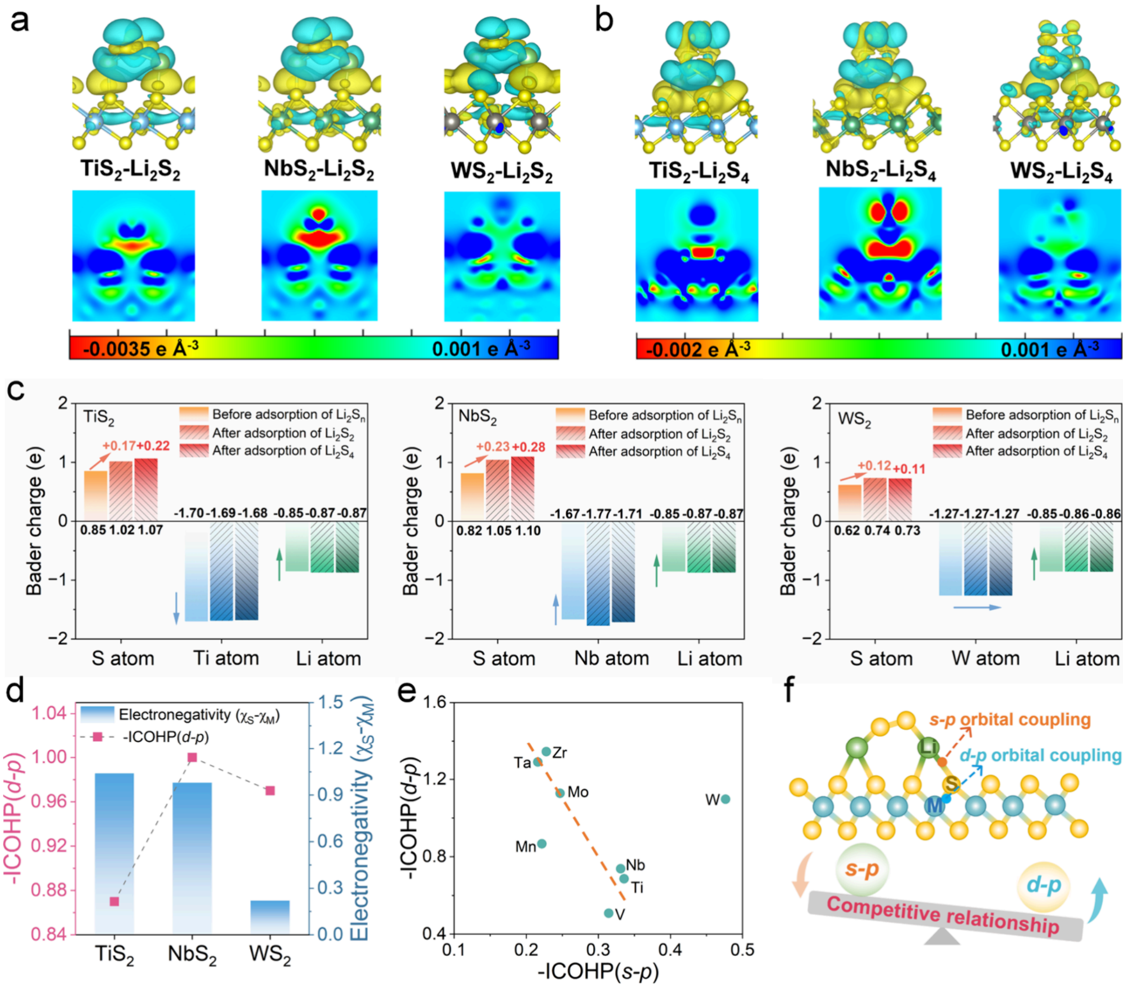

Figure 4. Electronic structure origin of TMS on the catalytic activity. Differential charge and 2D charge density maps of TiS_2, NbS_2, and WS_2 upon adsorption of (a) Li_2S_2 and (b) Li_2S_4. The yellow and blue regions represent charge accumulation and charge depletion (isosurface value: 0.02 e Å^{-3}), respectively. (c) Charge transfer of TiS_2, NbS_2, and WS_2 before and after the adsorption of Li_2S_2 and Li_2S_4. (d) Relationship between the M−S electronegativity difference and M−S bond strength (ICOHP) for TiS_2, NbS_2, and WS_2. (e) Correlation between the bonding strength (−ICOHP) of subsurface metal-surface sulfur (M−S) bonds and that of Li−S interactions formed between LiPSs and the TMS surface. (f) Competitive relationship between *d-p* orbital coupling and *s-p* orbital coupling.

effectively reducing reaction overpotentials and enhancing catalytic performance (Figure 3e). This pathway-dependent descriptor circumvented the intrinsic limitations imposed by the scaling relationships, underscoring the importance of minimizing the energy barrier differences between these critical conversion steps.

Electronic Structure Origin of the Catalytic Activity

To show the origin of the pathway-dependent descriptor and catalytic activity of TMSs, we analyzed the electronic structure of Li_2S_4 and Li_2S_2 adsorbed on the TMS surface (Figures 4a,b and S10). The blue regions around the Li atoms indicate electron loss, while the yellow regions near the S atoms in the TMS indicate electron gain. Meanwhile, the presence of fewer yellow and blue regions between atoms in Li_2S_2 and Li_2S_4 suggests that charge redistribution has occurred within themselves. The two-dimensional charge density maps of three representative TMSs (M = Ti, Nb, W) from different periods and groups further illustrate that the largest amount of electron transfer occurs between NbS_2 and Li_2S_2 or Li_2S_4, followed by TiS_2, with WS_2 showing the least. This indicates that NbS_2 exhibits the strongest electronic interactions with LiPSs, which facilitates enhanced reaction kinetics.[46,47] Bader charge analysis quantitatively assessed the charge transfer. Before Li_2S_2 or Li_2S_4 adsorption, the surface sulfur atoms of TMSs gain electrons from subsurface transition-metal atoms.[48] The greater the electronegativity difference between the sulfur and transition metal atoms (TiS_2 > NbS_2 > WS_2), the more electrons acquired by the sulfur atoms (Figure 4c,d). The electron numbers gained by sulfur atoms in TiS_2, NbS_2, and WS_2 are 0.85, 0.82, and 0.62 e, respectively (Figure 4c). After the adsorption of Li_2S_2 and Li_2S_4, the number of electrons gained by the S atoms on the catalyst surface was further increased due to the transfer of electrons from the surface-adsorbed Li atoms to the S atoms, in addition to the electron supply from the transition metal atoms. In the three TMS systems, the electron loss by Li atoms shows a consistent trend before and after LiPS adsorption with minimal differences. Therefore, the main variation in the electron gain by sulfur atoms is attributed to the electrons provided by the transition metal atoms. Specifically, the number of electrons supplied by Ti atoms to surface sulfur atoms decreases after LiPS adsorption (from 1.70 to 1.68 e), while the electron contribution from Nb atoms increases significantly (from 1.67 to 1.71 e), and that from W atoms remains 1.27 e nearly

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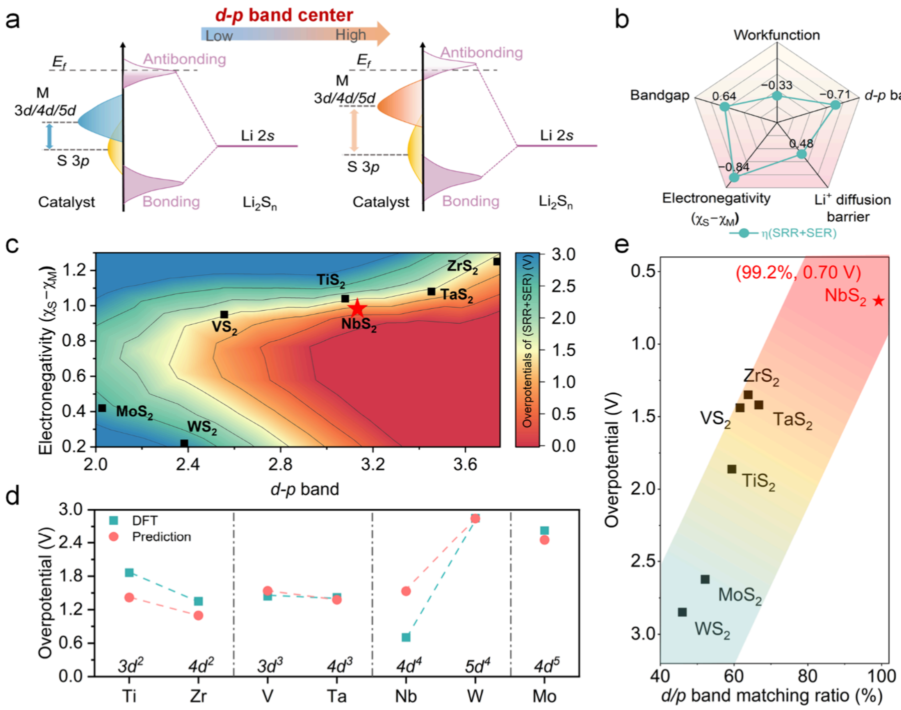

Figure 5. Electronic descriptors highly correlated with TMS catalytic activity and theoretical model predictions. (a) The effect of the *d-p* band center of TMS on the bonding between Li atoms in LiPSs and sulfur atoms on the TMS surface. (b) Pearson correlation analysis between five electronic structural features and the theoretical overpotential (the sum of SRR and SER overpotentials). (c) The relationship of the electronegativity difference ($\chi_S - \chi_M$) and *d-p* band center of TMS on theoretical overpotential. (d) Consistency between theoretical model predictions and DFT calculation results. (e) Relationship between the theoretical overpotential of TMS and the corresponding *d-p* band-matching percentage.

unchanged. Among the investigated catalysts, NbS_2 possesses the most favorable electronic structure for facilitating charge redistribution, as reflected by its strongest electronic interaction with LiPSs and the significant increase in electron donation from Nb atoms after adsorption.

Crystal orbital Hamilton population (COHP) analysis (Figure S11) shows that TMS predominantly exist in bonding states below the Fermi level; the strength of the M−S bond is further quantified by the integrated crystal orbital Hamilton population (ICOHP) value. A higher −ICOHP value corresponds to a stronger M−S bonding strength.[37] The Nb−S bond exhibits the highest bonding strength (−1.00), followed by the W−S bond (−0.97), while the Ti−S bond has the weakest bond strength (−0.87) (Figure 4d). This explains that after the adsorption of LiPSs, Nb atoms are still able to donate electrons to S atoms, while Ti atoms exhibit a decreased electron donation capacity. The electron transfer behavior between the catalyst and polysulfides is consistent during the adsorption of Li_2S_4 and Li_2S_2.

Beyond evaluating intrinsic M−S bonding in TMS, we quantified Li−S interactions between LiPSs and the TMS surfaces. ICOHP analysis reveals an inverse correlation between M−S and Li−S bond strengths postadsorption (Figure 4e, S12 and S13). Enhanced M−S bonding weakens Li−S interactions, impeding Li−S electron transfer and slowing reaction kinetics. The negative correlation between the M−S and Li−S bonding strengths reflects a competitive relationship between d-p orbital coupling and *s-p* orbital coupling (Figure 4f), where excessive *d-p* orbital overlap suppresses *s-p* interactions. Optimal catalytic performance therefore requires balanced orbital coupling to prevent extreme sulfur species adsorption.

Correlation between Redox Kinetics and Band Matching

The competitive *d-p* and *s-p* orbital coupling regulates sulfur species adsorption and electron transfer by catalyst electronic modulation.[49,50] As shown in Figure 5a, hybridization between the s orbital of Li and the p orbital of *S* forms bonding and antibonding orbitals during LiPS adsorption. The relative positioning of the *d*-band center of the metal atom and the *p*-band center of the surface sulfur atom (*d-p* band center) critically determines the coupling strength. Close *d-p* band center proximity strengthens *d-p* interactions but suppresses the *s-p* orbital coupling between Li and S through the competitive mechanism, weakening adsorption, hindering the suppression of the shuttle effect and slowing kinetics. Moderate *d-p* separation optimizes *s-p* hybridization, enhancing Li−S bonding to boost electron transfer and reduce reaction barriers. Excessive separation, however, triggers overadsorption that poisons catalytic sites, demonstrating the necessity for balanced *d-p* band engineering to optimize adsorption strength and catalytic efficiency.

To identify intrinsic electronic structure descriptors that are directly related to catalytic activity, conductivity, lithium-ion conductivity, work function, *d-p* band center, and M−S electronegativity difference were selected as features to correlate with the overpotential of the sulfur conversion reaction, the sum of the overpotential for the SRR and SER (η) was set as the target value (Figure S14 and Table S1). Based on the five features, we calculated the Pearson correlation coefficient between each feature and the overpotential (η; Figure 5b). The result shows that the reaction overpotential is influenced by multiple features, and thus, a single feature is inadequate for effectively screening and predicting high-

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performance catalysts. The Pearson correlation coefficient between the M−S electronegativity difference and η is −0.84, while that for the *d-p* band center is −0.71. These two parameters were chosen as the intrinsic electronic structure descriptors because they have the most significant influence on the catalytic activity. Figure 5c shows that under the synergistic influence of these two electronic structural parameters, NbS_2 exhibits the lowest reaction overpotential, demonstrating the optimal catalytic activity. A predictive equation for catalytic activity was then constructed using multiple linear regression:

$$\eta = k_1 \times (d - p \text{ band center}) - k_2 \times (\chi_S - \chi_M) + k_3$$

where k_1, k_2, and k_3 are 0.087, 1.811, and 3.034, respectively. The R^2 value of the model is 0.712.

The model indicates that a larger $\chi_S - \chi_M$ in TMS allows sulfur atoms to acquire more electrons from the transition metal atoms, placing the sulfur atoms in an electron-rich state. This electron-rich sulfur is more likely to adsorb Li^+ from polysulfide anions, thereby enhancing the anchoring capacity of the LiPSs. Meanwhile, an appropriate increase in the *d-p* band center strengthens the Li−S bonding, promotes interfacial electron transfer, and lowers the energy barrier of the sulfur conversion reaction, thereby significantly improving the catalytic activity.

To validate the accuracy of the model, we compared the predicted overpotential values with those obtained from the DFT calculations. Since the catalytic activity of the catalysts is strongly related to the *d* orbitals in their valence electron configuration, we classified the catalysts based on their *d*-electron number. Figure 5d shows a high degree of consistency between the predicted values and the DFT-calculated values, confirming the reliability of the model in predicting the catalytic activity of the catalysts.[51] However, owing to the limited size of the current data set, the predictive performance of the model still has room for further improvement. Therefore, expanding the data set to improve the model's applicability and generalization ability will be an important direction for future research. The incorporation of more diverse and representative data is expected to further increase the R^2 value of the model, thereby enabling more accurate and robust predictions of the catalytic activity.

These results demonstrate that catalytic activity is jointly governed by the *d-p* band center position, the M−S electronegativity difference, and the valence electron configuration of the metal atoms, reflecting variations in the coupling strength between the metal *d* orbitals and sulfur *p* orbitals. Based on a systematic statistical analysis of the calculated results for all TMS systems, we introduce the concept of *d*/*p* band matching to quantitatively capture the influence of the electronic structure on catalytic activity. Specifically, our statistical analysis reveals a well-defined quadratic correlation between the *d-p* band center energy difference and the corresponding overpotential (Figure S1). Together with the statistically identified optimal *d*-orbital period number of 4 and the valence electron number of 5, these three key electronic-structure descriptors are used to construct the band-matching ratio. The optimal catalytic activity is achieved when the *d-p* band center energy difference approaches 3.11 eV. Therefore, the band-matching ratio reflects the extent to which the electronic structure of a given TMS approaches this optimal matching state. A higher band-matching ratio indicates a closer match to the optimal electronic configuration and is generally associated with a lower bifunctional overpotential. The detailed calculation formula is provided in Supplementary Note 5. As shown in Figure 5e, increasing the *d*/*p* band-matching ratio from 46.0% for WS_2 to 52.2% for MoS_2, 59.4% for TiS_2, and 63.8% for ZrS_2 leads to a pronounced decrease in the bifunctional overpotential from 2.84 to 1.35 V. NbS_2 exhibits the highest band-matching ratio (99.2%) and delivers the lowest overpotential (0.703 V). This behavior is consistent with the lattice-matching effect previously identified, indicating that improved electronic or geometric matching strengthens reactant−active-site interactions and accelerates sulfur conversion kinetics.[34]

Based on this quantitative relationship, it can be predicted that when the energy difference between the metal *d*-band center and the surface sulfur *p*-band center is 3.11 eV, the *d*-orbital period number is 4, and the valence electron count is 5, the system achieves 100% *d*/*p* band-matching, corresponding to a lower bound of 0.701 V for the bifunctional overpotential. This value defines the intrinsic performance ceiling of sulfur redox catalysis in pristine TMSs. Notably, this descriptor is not limited to layered TMS systems. When extended to a representative nonlayered sulfide (Ni_3S_2), it yields an overpotential (η) of 1.76 V with a corresponding *d*/*p* band-matching ratio of 68.5%, which falls well within the established linear correlation between the *d*/*p* band-matching ratio and overpotential (Figure S15). This observation indicates that the relationship holds across different crystal structures and coordination environments. Compared with previously proposed descriptors, our *d*/*p* band-matching descriptor combines the *d-p* band center, *d*-orbital period, and valence electron. These three factors jointly determine, from the perspectives of both covalency and ionicity, the orbital coupling strength between transition-metal compounds and adsorbed intermediates as well as the charge-transfer capability and therefore constitute the fundamental electronic-structure factors governing catalytic activity. Therefore, this concept can, in principle, be extended to other classes of sulfur hosts or even broader material systems, such as metal carbides, nitrides, and single-atom catalysts, where analogous orbital hybridization mechanisms play a critical role in catalytic processes.[52,53]

Experimental Validation of the Catalyst Performance

TiS_2, NbS_2, and WS_2 were typical catalysts to verify catalytic enhancements from breaking the scaling relationships. To ensure structural consistency between the experimental catalysts and theoretical models, comprehensive characterizations were conducted. TiS_2, NbS_2, and WS_2 exhibited typical layered nanosheet morphologies, as confirmed by scanning electron microscopy (SEM) (Figure S16). X-ray diffraction (XRD) analysis further verified the crystallographic phases of the materials: TiS_2 and NbS_2 exhibit the thermodynamically stable 1T phase, while WS_2 adopts the 2H phase, which is in full agreement with the DFT-selected crystal structures (Figure S17). In addition, electrical conductivity measurements were performed, showing that NbS_2 possessed the highest conductivity (201.91 S cm^{-1}), followed by TiS_2 (6.14 S cm^{-1}) and WS_2 (0.086 S cm^{-1}), in agreement with the trend of the calculated band gaps shown in Figure S3. These results collectively validate the suitability of the DFT structural models and the reliability of subsequent theoretical and kinetic analyses. Systematic assessments of LiPS interactions focused on the adsorption strength, lithium-ion migration dynamics, and the redox kinetics of the rate-determining step. Ultraviolet−visible (UV−vis) adsorption spectroscopy (Figure 6a)

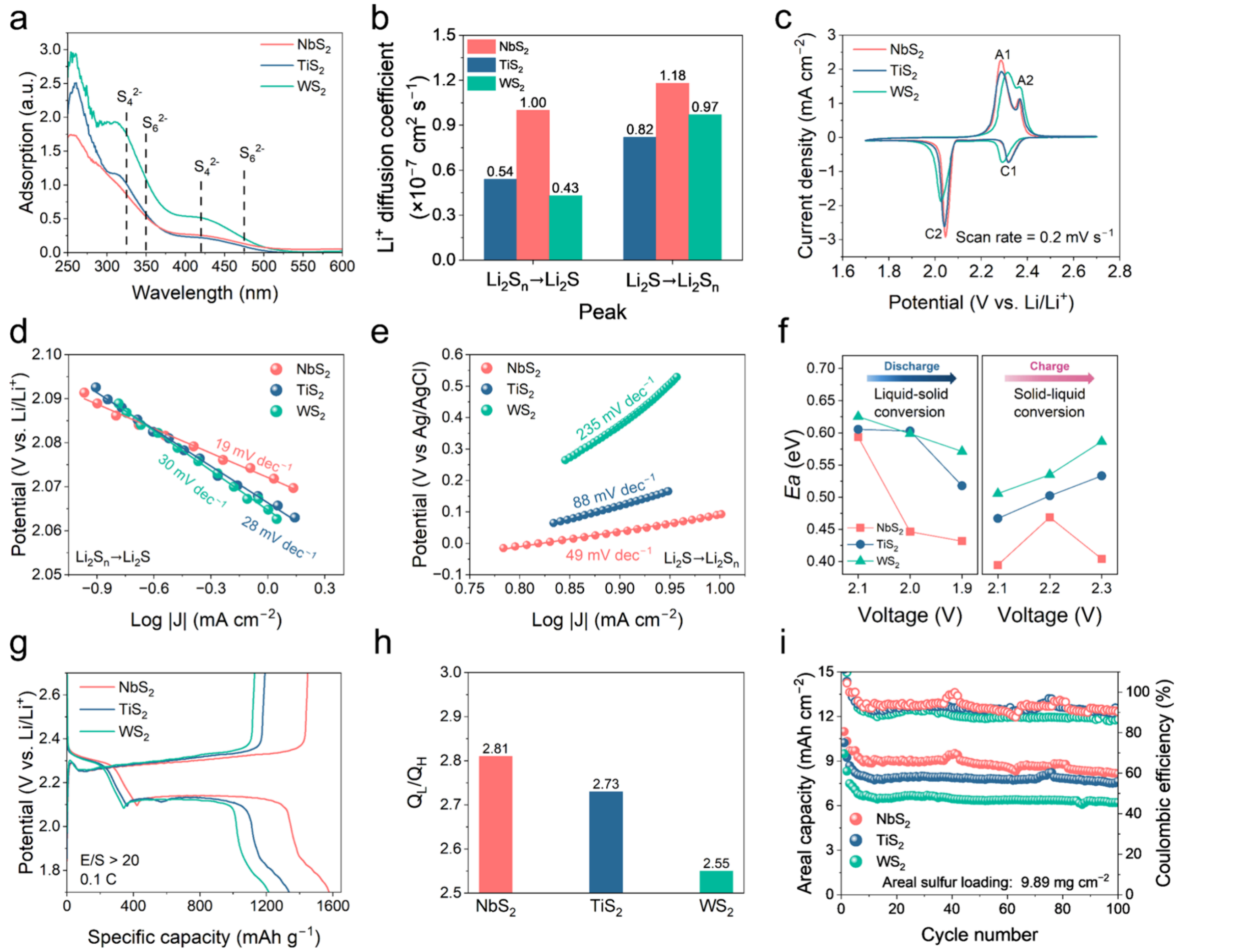

Figure 6. Reaction kinetics analysis and electrochemical performance of Li−S batteries containing TMSs. (a) UV−vis spectra of Li_2S_6 solution with TiS_2, Nb S_2, and WS_2. (b) Lithium-ion diffusion coefficient of TiS_2, NbS_2, and WS. (c) Cyclic voltammetry curves of lithium−sulfur full batteries containing TiS_2, NbS_2, and WS_2. (d) The fitted Tafel plots corresponding to the reduction peak from LiPSs to Li_2S in CV curves. (e) The fitted Tafel plots corresponding to LSV curves. (f) Activation energies of TiS_2, NbS_2 and WS_2 at given voltages. (g) Galvanostatic discharge−charge profiles at 0.1 C with TiS_2, NbS_2, and WS_2. (h) Capacity ratio of the low-voltage region and the high-voltage region (Q_L/Q_H). (i) Cycling stability of the batteries with high-sulfur loading at 0.1 C.

shows that compared to TiS_2 and WS_2, the NbS_2 in 0.01 M Li_2S_6 (DME/DOL = 1:1) solution exhibited a significantly reduced absorbance intensity at 420 and 460 nm,[54] corresponding to the characteristic peaks of S_4^{2-} and S_6^{2-}. This result suggests a stronger interaction between NbS_2 and LiPSs. This pronounced adsorption behavior can be attributed to the enhanced orbital hybridization between S 3*p* orbitals and the Li 2*s* orbitals in LiPSs. The lithium-ion diffusion coefficient (D_{Li^+}) was then calculated using the Randles-Sevcik equation based on cyclic voltammetry at varying scan rates (Figure 6b). A linear relationship between the peak current and the square root of the scan rate confirmed diffusion-controlled kinetics in these catalysts (Figures S18 and S19). Compared with WS_2 and TiS_2, NbS_2 showed the highest D_{Li^+} and best Li^+ transport capability, which is consistent with the CI-NEB calculation result.

Subsequently, we conducted a series of kinetic tests to establish a specific correlation between sulfur conversion performance and the pathway-dependent descriptor. The cyclic voltammetry (CV) curves show two distinct redox peaks (Figure 6c). The two cathodic peaks at ∼2.3 V (C1) and ∼2.1 V (C2) correspond to the reduction of elemental sulfur to long-chain LiPSs and their further conversion to Li_2S, respectively, while the two anodic peaks at ∼2.3 V (A1) and ∼2.4 V (A2) are associated with the reverse oxidation of Li_2S and LiPSs. Compared to TiS_2 and WS_2, the cathodic peak of NbS_2 shifted significantly toward higher potential, while the anodic peak shifted toward lower potential, with higher peak currents, indicating a significant enhancement in sulfur redox kinetics. The Tafel slope fitted from the second cathodic peak (C2) was as low as 19 mV dec^{-1} for NbS_2, which is substantially lower than that of TiS_2 (28 mV dec^{-1}) and WS_2 (30 mV dec^{-1}), further confirming its superior kinetic performance (Figure 6d). The nucleation-to-transformation ratio (NTR) was employed as a semiquantitative descriptor, calculated by comparing the integrated CV charges corresponding to the formation of Li_2S_4 at the C1 peak and the subsequent conversion to Li_2S at the C2 peak. A higher NTR value indicates a more complete utilization of Li_2S_4 and more efficient liquid-to-solid phase transformation. NbS_2 exhibited

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consistently higher NTR values over a range of scan rates, indicating its superior capability to promote efficient LiPSs conversion (Figure S20).[55] Furthermore, linear sweep voltammetry (LSV) measurements were conducted in electrolytes containing either 0.2 M Li_2S_4 or 0.2 M Li_2S_2 (Figure S21) to decouple the catalytic contributions of TMS toward the conversion of Li_2S_4 and Li_2S_2. In both cases, NbS_2 exhibits the highest peak current, demonstrating its excellent catalytic activity toward the two corresponding transformation pathways, which is consistent with the catalytic activity described by the $\Delta G_{Li_2S_4} - \Delta G_{Li_2S_2}$ and $\Delta G_{Li_2S_2} - \Delta G_{Li_2S_4}$ in the linear relationship.

The potentiostatic nucleation experiment with Li_2S further validated the role of TMS in regulating Li_2S conversion kinetics. NbS_2 exhibited the highest deposition specific capacity (576.98 mAh g^{-1}) compared with TiS_2 (512.78 mAh g^{-1}) and WS_2 (332.54 mAh g^{-1}), indicating its effective promotion of Li_2S nucleation and growth (Figure S22). Based on our calculated Li_2S adsorption energies, TiS_2 possesses the strongest Li_2S binding energy among the three catalysts (TiS_2 > NbS_2 > WS_2), which may impede the dynamic rearrangement and growth of Li_2S nuclei due to overly strong surface adsorption. In contrast, the moderate binding strength of NbS_2 appears to be more favorable for balancing nucleation and subsequent growth during the deposition process. Guided by the previously established linear relationship, which links the catalytic activity toward both Li_2S formation and decomposition with reaction energetics, we further evaluated the dissociation kinetics of Li_2S through LSV analysis. NbS_2 exhibited a significantly higher onset potential than TiS_2 and WS_2 (Figure S23). Tafel plots were derived by fitting the initial region of the oxidation peak in the LSV curves (Figure 6e). The Tafel slopes of NbS_2, TiS_2, and WS_2 are 49, 88, and 235 mV dec^{-1}, respectively, further confirming the kinetic advantage of NbS_2 in accelerating the sulfur evolution reaction. To comprehensively investigate the changes in the activation energy (E_a) during the sulfur redox process, temperature-dependent electrochemical impedance spectroscopy (EIS) measurements were conducted at various potentials. Nyquist plots recorded at 10, 20, 30, 40, and 50 °C were fitted using an equivalent circuit model to extract the charge-transfer resistance (R_{ct}) (Figure S24). The E_a for both the SRR and SER was calculated based on the Arrhenius equation. For the SRR, particularly in the liquid-to-solid phase transformation (LiPSs to Li_2S_2 or Li_2S), a relatively high E_a was observed due to the phase transformation resistance. Within this potential range (2.1−1.9 V), NbS_2 exhibited significantly lower E_a compared to TiS_2 and WS_2.[46] In contrast, during the solid-to-liquid phase transition in the SER, the inherently insulating nature of Li_2S results in a high oxidative energy barrier. Notably, the E_a for NbS_2 decreased to 0.40 eV at 2.3 V, suggesting that the comproportionation reaction facilitates the oxidation of Li_2S. Overall, NbS_2 consistently exhibits the lowest E_a across the entire potential range, indicating its strong capability to lower reaction energy barriers and accelerate sulfur redox kinetics. TiS_2 shows middle activation energy, while WS_2 displays the highest, in good agreement with the theoretical predictions.[56,57]

The electrochemical performance of Li−S cells using NbS_2, TiS_2, and WS_2 as sulfur cathode catalysts was evaluated to elucidate the catalytic effects of TMSs on sulfur redox reactions (catalyst content in cathode: 3%, sulfur mass loading: ∼1 mg cm^{-2}). As shown in Figure 6g, the cell with NbS_2 exhibited the highest specific capacity. The high voltage plateau corresponds to the reduction of elemental sulfur to LiPSs, while the low voltage plateau is associated with the further conversion of LiPSs to Li_2S. The capacity ratio between the two plateaus (Q_L/Q_H) reflects the extent of sulfur utilization. The Q_L/Q_H value of NbS_2 is the highest at 2.81, followed by TiS_2 (2.73), and WS_2 showed the lowest value of 2.55 (Figure 6h). Moreover, the voltage polarization between the charge and discharge plateaus of NbS_2 is lower (157 mV) than that of TiS_2 (178 mV) and WS_2 (181 mV). The reduced polarization of the NbS_2-based cathode can be primarily attributed to the lower kinetic barriers for both Li_2S_n reduction and Li_2S oxidation, which is consistent with theoretical predictions. The NbS_2 cathode also has excellent rate performance (Figure S25). Even at a high rate of 3.0 C, it maintained a specific capacity of 678 mAh g^{-1}, which is higher than that of TiS_2 (620 mAh g^{-1}) and WS_2 (445 mAh g^{-1}). When the rate was returned from 3.0 to 1.0 C, the capacity nearly recovered to its initial value, indicating excellent rate reversibility. In addition, in situ XRD measurements show that the characteristic peaks of the 1T-NbS_2 phase remain stable during discharge and charge (Figure S26), indicating its good structural stability during the electrochemical process. The cell with NbS_2 maintained a capacity of 470 mAh g^{-1} after 1000 cycles at 1.0 C, with a minimal capacity decay rate of 0.086% per cycle, outperforming TiS_2 (400 mAh g^{-1}, 0.23%) and WS_2 (333 mAh g^{-1}, 0.18%) (Figure S27). Furthermore, at a high areal sulfur loading of 9.89 mg cm^{-2}, the cell with NbS_2 delivers an initial areal capacity of 10.98 mAh cm^{-2} at 0.1 C, and it still retains 8.13 mAh cm^{-2} after 100 cycles, corresponding to 90.29% capacity retention. This performance significantly outperforms the cells with TiS_2 and WS_2 cathodes (Figure 6i) and exceeds that of most reported TMS-based cathode systems for Li−S batteries, further validating the effectiveness of the NbS_2 catalyst screened by the *d*/*p* band-matching descriptor (Table S5).[56,57] Moreover, a multilayer pouch cell with a sulfur loading of 5.1 mg cm^{-2} was assembled, delivering a total capacity of 2.16 Ah and an initial specific energy of 401 Wh kg^{-1} at 129.1 mA, while also exhibiting stable charge/discharge profiles and reversible cycling over about 20 cycles with high Coulombic efficiency (Figure S28).

■ CONCLUSION

In summary, this work establishes *d*/*p* band-matching as a predictive electronic-structure descriptor that overcomes the conventional adsorption−activity scaling relations limiting multistep sulfur redox kinetics in transition-metal sulfide (TMS) model catalysts. By jointly considering the intrinsic electronic properties of the transition metal and the degree of *d*−*p* band alignment between metal and surface sulfur, a linear relationship between the band-matching ratio and the overpotentials of sulfur reduction and evolution reactions is uncovered, enabling quantitative activity prediction beyond volcano-type behavior. Engineering the band-matching ratio drives surface sulfur atoms into an electron-rich state, strengthens orbital coupling with lithium polysulfides, and lowers the kinetic barriers for sulfur conversion. Among TMSs, NbS_2 achieves an exceptionally high band-matching ratio of 99.2%, delivering the lowest bifunctional overpotential (0.70 V), 4-fold lower than WS_2 with a band-matching ratio of 46.0%. The Li−S battery employing NbS_2 enables the lowest electrochemical barriers for LiPS reduction and oxidation and

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exhibits superior cycling stability, even at a high sulfur loading of 9.89 mg cm^{-2}. Beyond identifying an optimal TMS catalyst, this work defines an intrinsic performance ceiling for sulfur redox catalysis in pristine TMSs, providing a general electronic structure design principle for tuning the catalytic activity in Li−S batteries. The concept of *d*/*p* band-matching further offers a transferable framework for the rational design of high-performance catalysts in broader electrochemical energy-conversion reactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.6c02040.

Details of materials, chemicals, and methods used in this work, including the DFT calculations, Gibbs free energy calculations, Pearson correlation analysis, preparation of the sulfur cathode, electrochemical measurements and materials characterizations, as well as supplementary figures and tables (PDF)

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Notes

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